

GAUTAM RAJ

AI/ML Developer | Full-Stack Engineer

gp832008@gmail.com | +91 8809339659 | linkedin.com/in/gautam-raj-039204294 | github.com/GautamRaj1234

PROFESSIONAL SUMMARY

Final-year B.Tech CSE (AI) student and full-stack/AI engineer who independently designs, builds, and ships production software end-to-end. Delivered PlantSathi AI, a full-stack plant-health platform combining computer vision, an LLM-powered conversational assistant, and live weather data (React, Node.js/Express, JWT auth, deployed on Vercel and Render), and SmarHire, a RAG-based resume-screening SaaS platform (FastAPI, React, ChromaDB, Google Gemini API) live on Render and Vercel. Completed two internships in front-end and full-stack web development, and holds 8 Cisco Networking Academy certifications. Seeking roles in AI/ML or full-stack development.

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++, SQL

AI / ML / NLP: Retrieval-Augmented Generation (RAG), LLM APIs (Google Gemini, Groq/Llama), Computer Vision & Image Classification APIs (PlantNet, Hugging Face Inference), Vector Embeddings & Semantic Search, TF-IDF, Cosine Similarity, NLTK, scikit-learn, NumPy, Pandas

Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST API Design, JWT Authentication, bcrypt, SQLAlchemy, Role-Based Access Control

Frontend: React.js, React Router, Vite, Tailwind CSS

Databases & Vector Stores: SQL, SQLite, PostgreSQL, ChromaDB, FAISS

DevOps & Tools: Git, GitHub, Docker, GitHub Actions (CI/CD), Render, Vercel, VS Code

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS

PROJECTS

PlantSathi AI — AI-Powered Plant Health & Care Platform

Live

React, Node.js, Express, JWT, PlantNet API, Hugging Face Inference, Groq (Llama 3.1), OpenWeatherMap | [GitHub: github.com/GautamRaj1234/plantsathi-ai](https://github.com/GautamRaj1234/plantsathi-ai) | [Live: plantsathi-ai.vercel.app](https://plantsathi-ai.vercel.app)

- Architected and independently shipped a full-stack plant-health platform end-to-end — React (Vite + Tailwind) frontend deployed on Vercel, Node.js/Express backend deployed on Render — with JWT authentication, bcrypt password hashing, and strict per-user data isolation enforced across every protected endpoint.
- Built a multi-model AI pipeline integrating three external AI services: PlantNet's computer-vision API for plant species identification, a Hugging Face vision model for leaf-disease classification, and Groq's Llama 3.1 for a conversational, context-aware AI plant-care assistant.
- Integrated live weather data (OpenWeatherMap) to generate location- and season-aware care recommendations, and built an analytics dashboard visualizing scan history, diagnosis outcomes, and plant-health trends over time.
- Diagnosed and resolved real production issues, including a breaking Hugging Face Inference API migration, a Vite dev-proxy bug causing intermittent multipart file-upload failures, and a subtle ES-module import-ordering bug that silently broke JWT session verification.
- Designed resilient error handling with automatic fallback to demo data whenever a third-party AI API is unavailable, keeping the app fully functional and demonstrable without requiring live API keys.

SmarHire — AI-Powered Resume Screening & Candidate Matching Platform

Live

Python, FastAPI, React, ChromaDB, Google Gemini API, Docker, JWT | [GitHub: github.com/GautamRaj1234/smarthire](https://github.com/GautamRaj1234/smarthire) | [Live: frontend-olive-one-16.vercel.app](https://frontend-olive-one-16.vercel.app)

- Architected and independently shipped a full-stack SaaS platform end-to-end — FastAPI backend, React (Vite + Tailwind) frontend — deployed live on Render (API) and Vercel (frontend) with CI/CD via GitHub Actions.
- Built a hybrid resume-matching engine combining semantic similarity (Gemini embeddings + ChromaDB vector search, cosine similarity) with keyword-based ATS scoring across a 60+ skill taxonomy, producing a weighted composite match score.
- Integrated Google Gemini for AI-generated candidate analysis — strengths/weaknesses, resume improvement suggestions, and three categories of tailored interview questions — with automatic fallback to keyword-only results if the API call fails.
- Implemented JWT authentication with role-based access control (recruiter vs. candidate), enforcing strict per-user data isolation across every API endpoint.
- Designed a recruiter dashboard with filterable/sortable candidate rankings, analytics visualizations, CSV export, and one-click PDF report generation; built a companion candidate-facing self-check tool for independent resume evaluation.
- Diagnosed and resolved real production issues — CORS misconfiguration, memory-constrained free-tier hosting (migrated a local ML embedding model to a hosted API to cut memory usage), and a breaking third-party model deprecation.

INTERNSHIP & EXPERIENCE

Front-End Web Developer Intern — Edunet Foundation (IBM SkillsBuild) Aug 2025 – Oct 2025

- Built 5+ responsive web interfaces using HTML, CSS, and JavaScript, improving accessibility and cross-browser consistency to meet modern UI/UX standards.
- Integrated 2 third-party REST APIs to enable real-time dynamic content rendering across pages.
- Contributed to 4 feature releases over 8-week Agile sprints, gaining practical exposure to React.js for component-based UI work.

Full Stack Web Development Intern — Soft Nexis Technology Mar 2026 – May 2026

- Contributed to front-end and back-end features using HTML, CSS, JavaScript, and modern web frameworks, applying Agile workflows throughout the engagement.
- Demonstrated strong coordination and attention to detail across team collaborations, delivering quality work across multiple project features.

Certificate No. SN1001133 — MSME-registered company (UDYAM-UP-17-0032005)

EDUCATION

B.Tech — Computer Science Engineering (AI) 2023 – 2027 (Expected)

Motihari College of Engineering, Motihari, Bihar | CGPA: 8.16

Senior Secondary (XII) 2021 – 2023

Dr. J. Mishra College, Muzaffarpur, Bihar | 79%

CERTIFICATIONS

- IBM SkillsBuild — Front End Web Development
- Electronics & ICT Academies (MeitY / IITs) — QT-03 Basics of Quantum Programming (40 hrs, 3 Credits), May–Jun 2025, Performance: Excellent
- Soft Nexis Technology — Full Stack Web Development Internship Certificate (SN1001133), Mar–May 2026
- Cisco Networking Academy — CCNA 1 (Introduction to Networks), CCNA 2 (Switching, Routing & Wireless Essentials), CCNA 3 (Enterprise Networking, Security & Automation)
- Cisco Networking Academy — Python Essentials 1 & 2; Introduction to Data Science; Introduction to Modern AI; Apply AI: Analyze Customer Reviews